META-LABELING LAB II

# Meta-Labeling in Practice

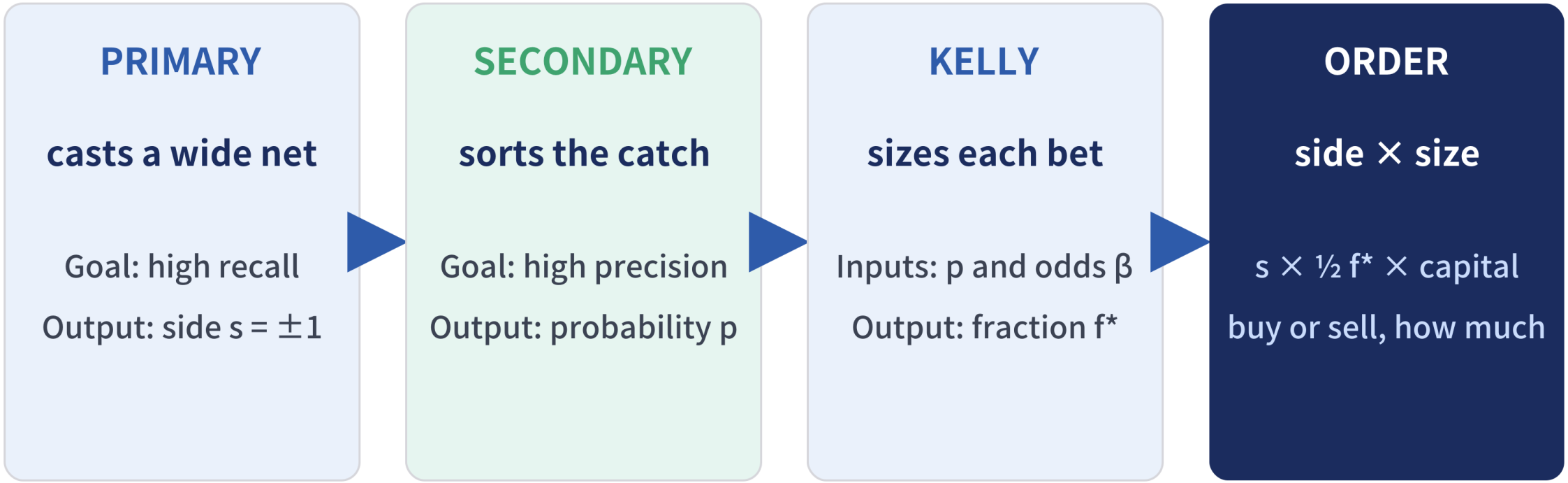
## Primary side → secondary probability → Kelly position

One SPY example, then 22 ETFs out of sample (2015–2026)

The primary casts a wide net for recall, a logistic model sorts the catch for precision, and the Kelly formula turns the probability into a position size.

# Three models: which way, whether, how much

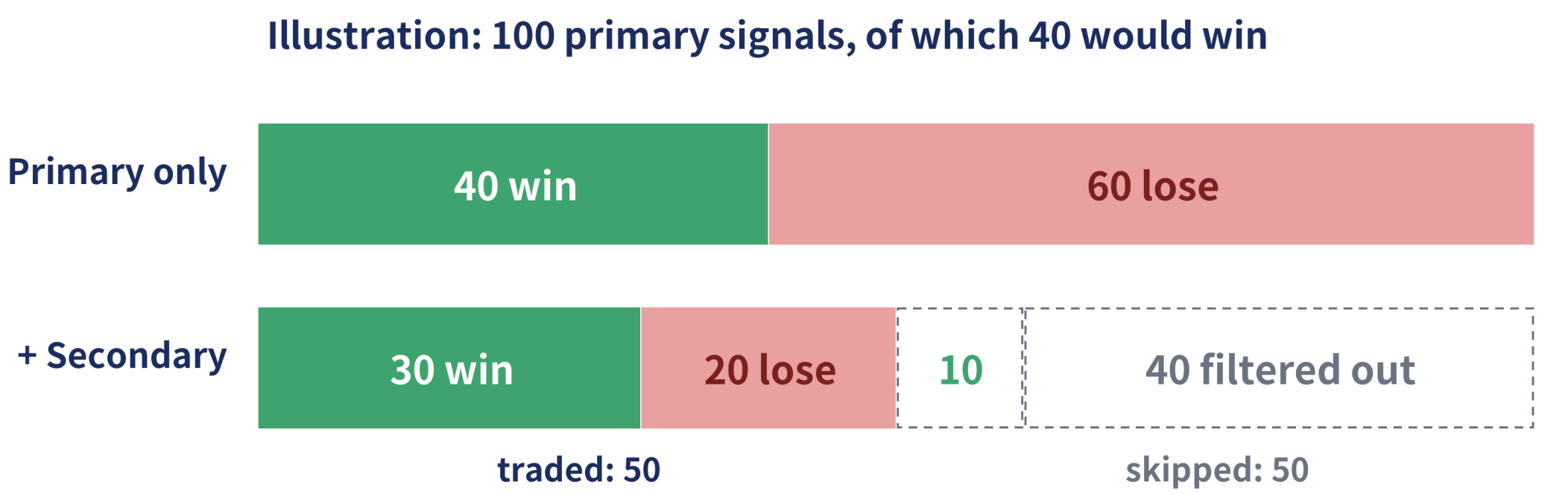
Two models with different goals feed one sizing rule (López de Prado, 2018)




**Primary: which way? Secondary: will it work? Kelly: how much?**

# The secondary trades a little recall for much precision

※ Illustrative numbers: the secondary keeps 50 of 100 signals and drops 40 of the 60 losers




**Precision 40% → 60%**
30 winners out of 50 trades

**Recall 100% → 75%**
30 of the 40 winners kept

# Precision and recall judge the two models

TP: traded and won · FP: traded and lost · FN: skipped but would have won

$$\text{precision} = \frac{TP}{TP+FP} \qquad \text{recall} = \frac{TP}{TP+FN}$$

## Primary: judged by recall

- Its job is not to miss a chance
- A loose rule is fine: many signals, many losers
- Here: fire whenever price leaves the ±2σ band

## Secondary: judged by precision

- Its job is to trade only signals likely to win
- It sees only the primary's signals
- Here: a logistic model gives each signal a p

01

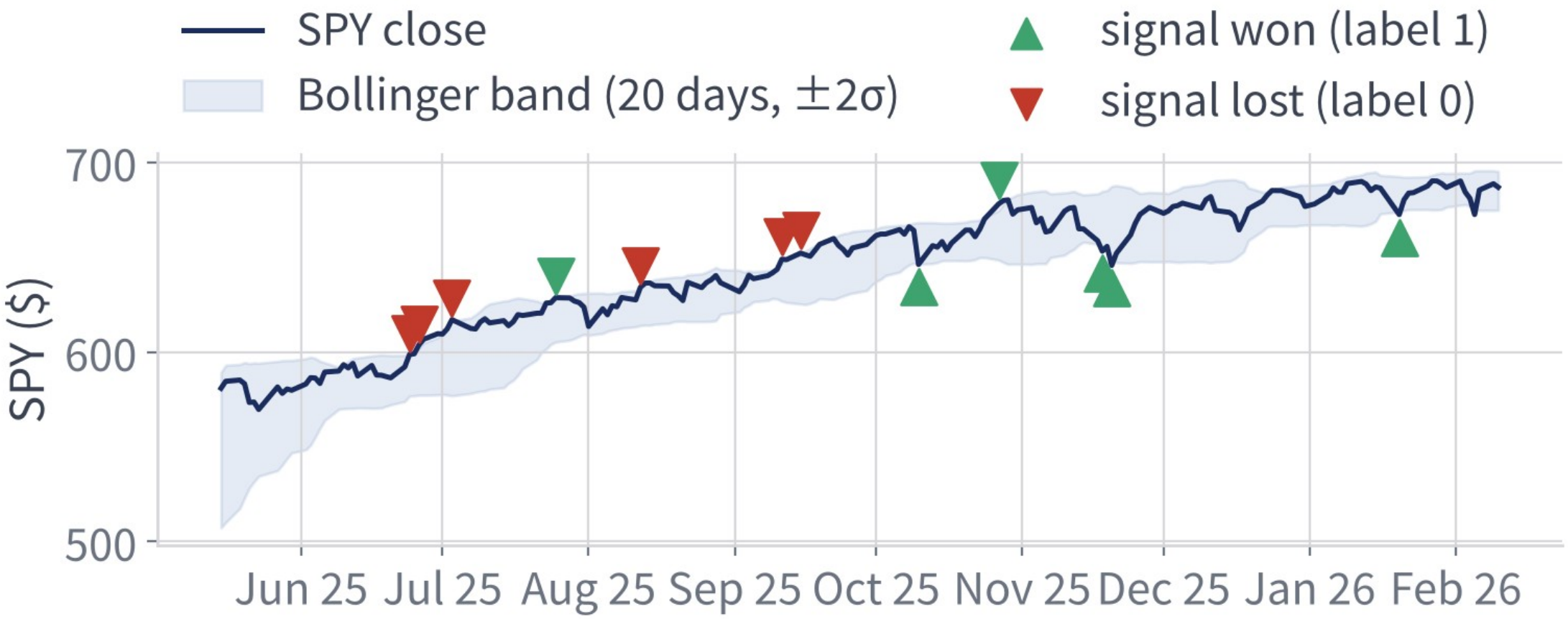
STEP 1 · PRIMARY

# The Primary Model Sets the Side

A simple Bollinger band rule — and how its hit ratio is counted

# The primary fires whenever price leaves the band

SPY, Jun 2025 – Jan 2026: buy below −2σ, sell above +2σ; 12 signals, green won and red lost



# The triple barrier decides whether each signal won

Barriers at ±1.5 × daily volatility with a 10-day limit; the first one touched decides




$$\text{meta-label} = \begin{cases} 1 & \text{if } s \times r > 0 \\ 0 & \text{otherwise} \end{cases}$$

## SPY buy, Nov 20 2025

- s = +1 (buy), r = return to the first barrier
- Profit line hit on Nov 24
- So the label is 1: the primary was right

# The hit ratio is the share of signals labeled 1

Count the labels: SPY's 12 signals won 6 times; all 22 ETFs won 1,927 of 3,721 times

## SPY, June 2025 – January 2026: the primary fired 12 times



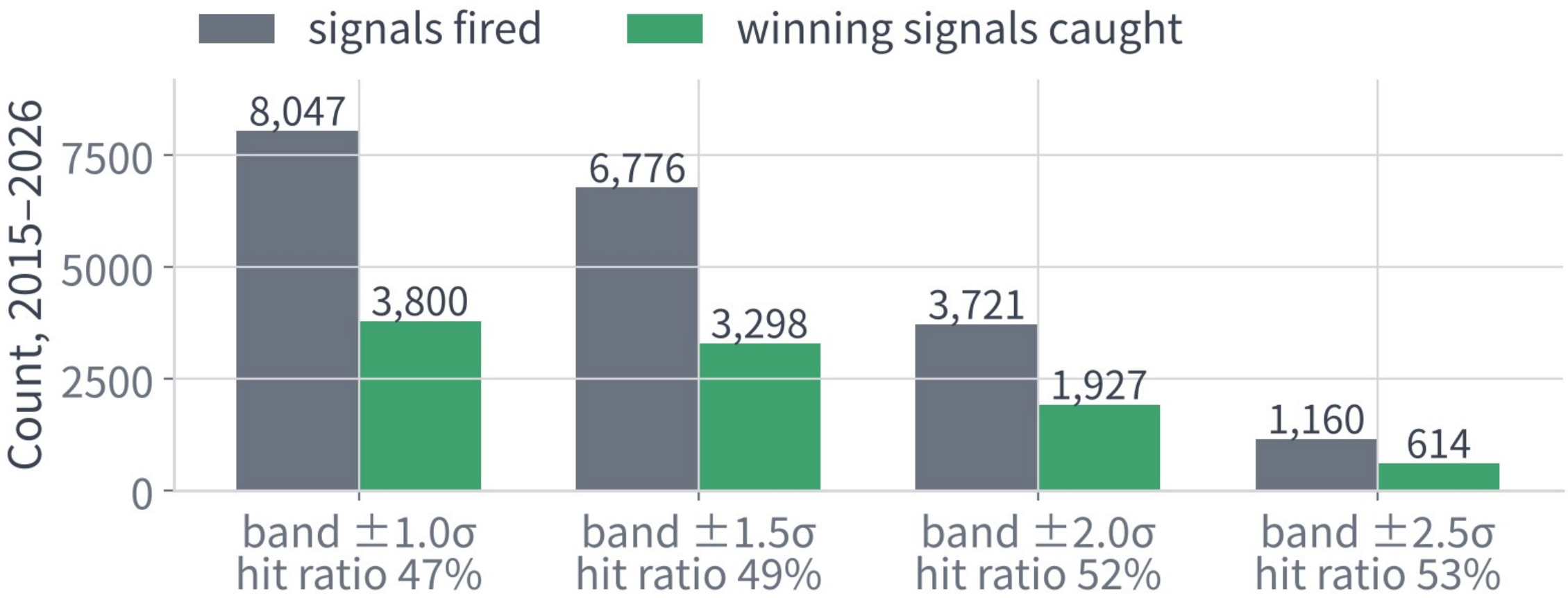

1 = the first barrier touched was profit-take (or the trade was up at the time limit); 0 = otherwise

**hit ratio = 6 wins / 12 signals = 50%**

full sample 2015–2026: 1,927 / 3,721 = 51.8%

# A wider band catches more winners at a lower hit ratio

The primary's knob sets recall: ±1σ catches 3,800 winners, ±2.5σ only 614 (22 ETFs, 2015–2026)



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STEP 2 · SECONDARY

# The Secondary Model Estimates p

A logistic model learns from the primary's own record when its signals tend to work

# The secondary learns from the primary's own record

Each primary signal becomes one row: two features measured that day, and its 0/1 label

| SPY signal | Side | x1  trend agreement | x2  fear (VIX ratio) | Label |
|---|---|---|---|---|
| Jun 24 2025 | sell | −5.2% | 0.73 | 0 |
| Jul 25 2025 | sell | −9.1% | 0.80 | 1 |
| Sep 11 2025 | sell | −10.6% | 0.90 | 0 |
| Oct 10 2025 | buy | +8.8% | 1.36 | 1 |
| Nov 20 2025 | buy | +6.6% | 1.50 | 1 |

### x1: trend agreement

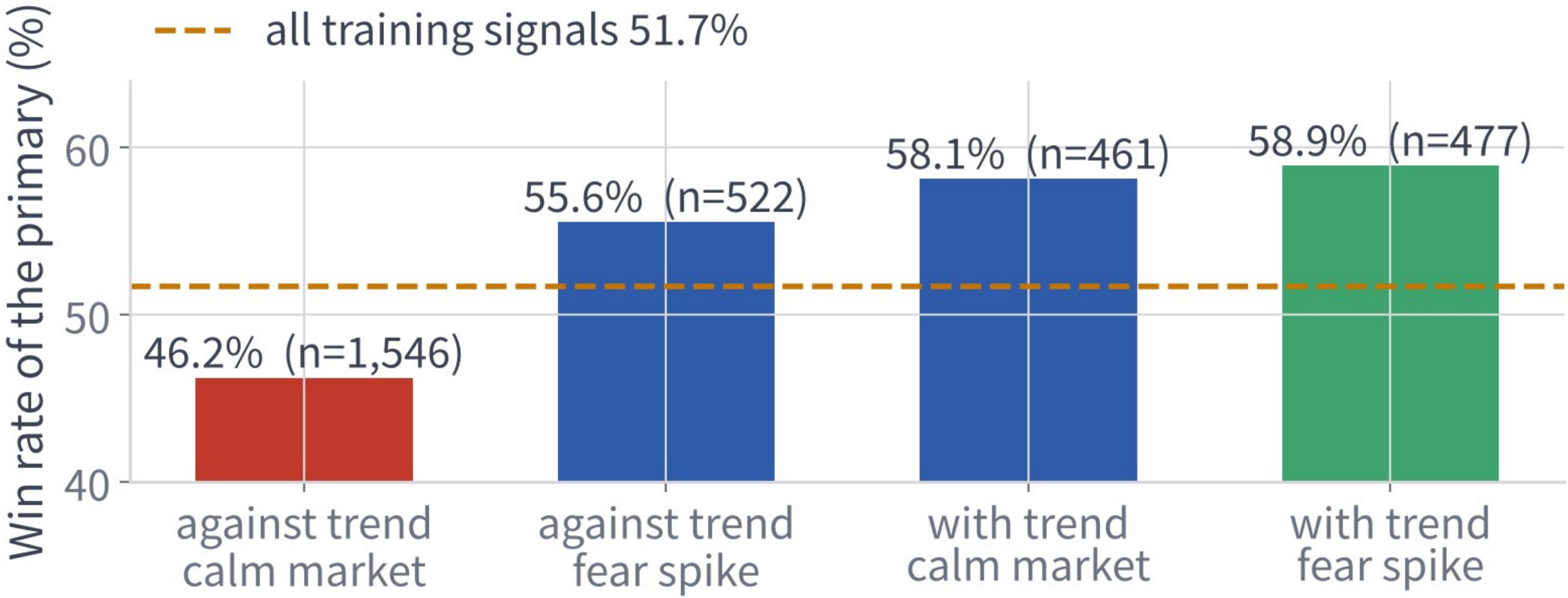
- % above the 200-day average, signed by the side

### x2: fear spike

- VIX divided by its own 60-day average

# Fades work best with the trend and in fear

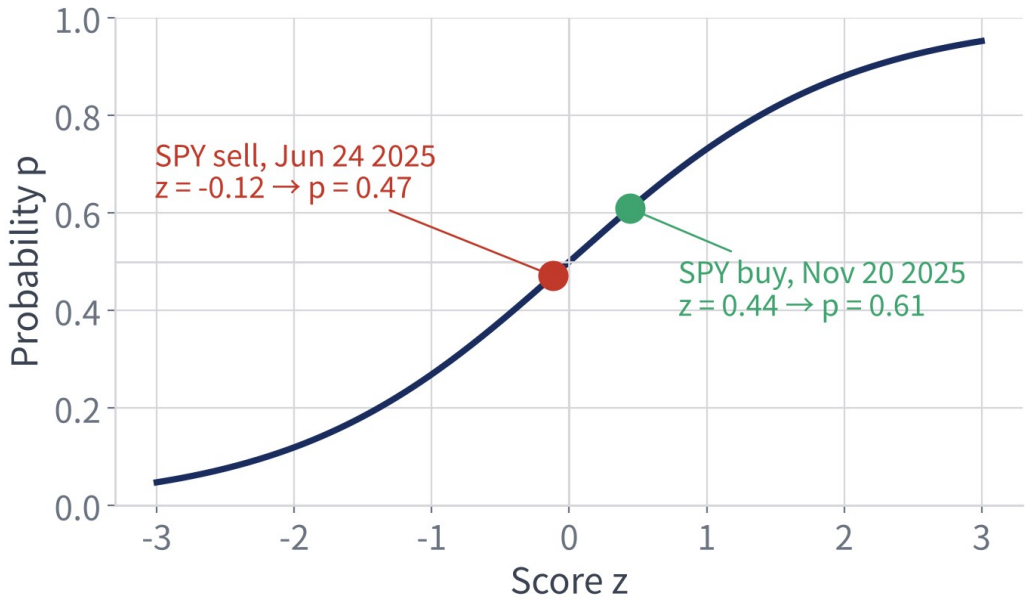
Training signals 2005–2014: primary win rate by trend agreement and VIX spike (above 1.1× normal)



# A logistic model turns the two features into p

Fitted on 3,006 training signals (2005–2014); the S-curve squeezes any score z into 0–1

$$z = \underbrace{-0.447}_{\text{base}} + \underbrace{0.012}_{\text{per 1\%}}\, x_1 + \underbrace{0.541}_{\text{per 1.0}}\, x_2 \qquad p = \frac{1}{1+e^{-z}}$$




## Reading the weights

- Each 1% of trend agreement adds 0.012 to z
- Each 1.0 of VIX ratio adds 0.541 to z
- Higher z gives higher p

# Two SPY signals pass through the model

Plug x1 and x2 into z, then z into the S-curve; trade only if p is above 0.55

$$z = -0.447 + 0.012 \times 6.62 + 0.541 \times 1.50 = 0.44 \quad \Rightarrow \quad p = \frac{1}{1+e^{-0.44}} = 0.61$$

$$z = -0.447 + 0.012 \times (-5.22) + 0.541 \times 0.73 = -0.12 \quad \Rightarrow \quad p = 0.47$$

### Buy, Nov 20 2025: p = 0.61

- With the trend, VIX 50% above normal
- p > 0.55, so trade: it won +2.5%

### Sell, Jun 24 2025: p = 0.47

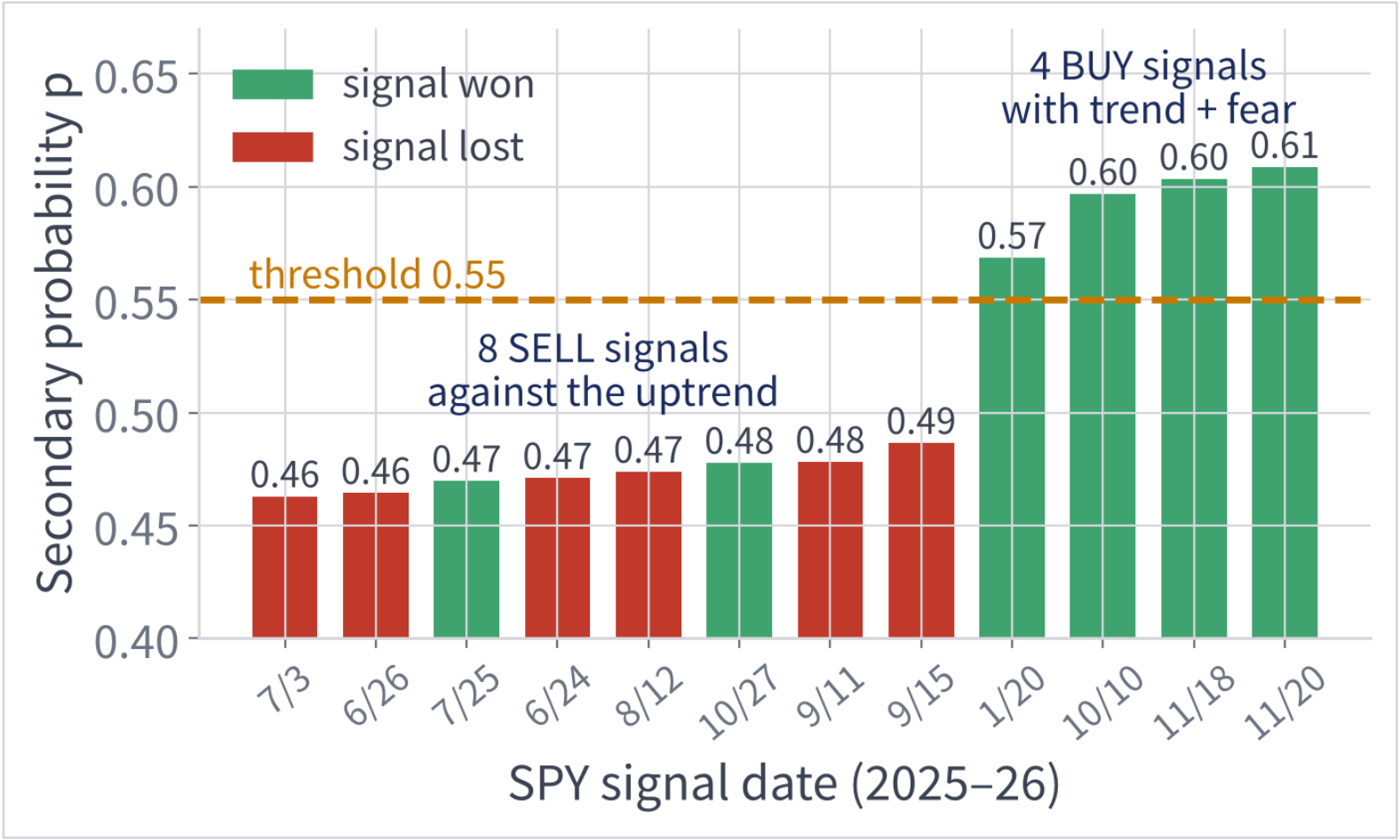
- Against the uptrend, calm market
- p < 0.55, so skip: it would have lost 2.2%

# On SPY the 0.55 line keeps four winners and no losers

The 12 SPY signals sorted by p: bars above the line are traded, bars below are skipped

## Count it

- Traded 4, won 4
- Precision 100%
- Winners in all: 6
- Recall 4/6 = 67%



# The same count, written as a confusion matrix

Rows: did the signal win? Columns: did the secondary trade it?

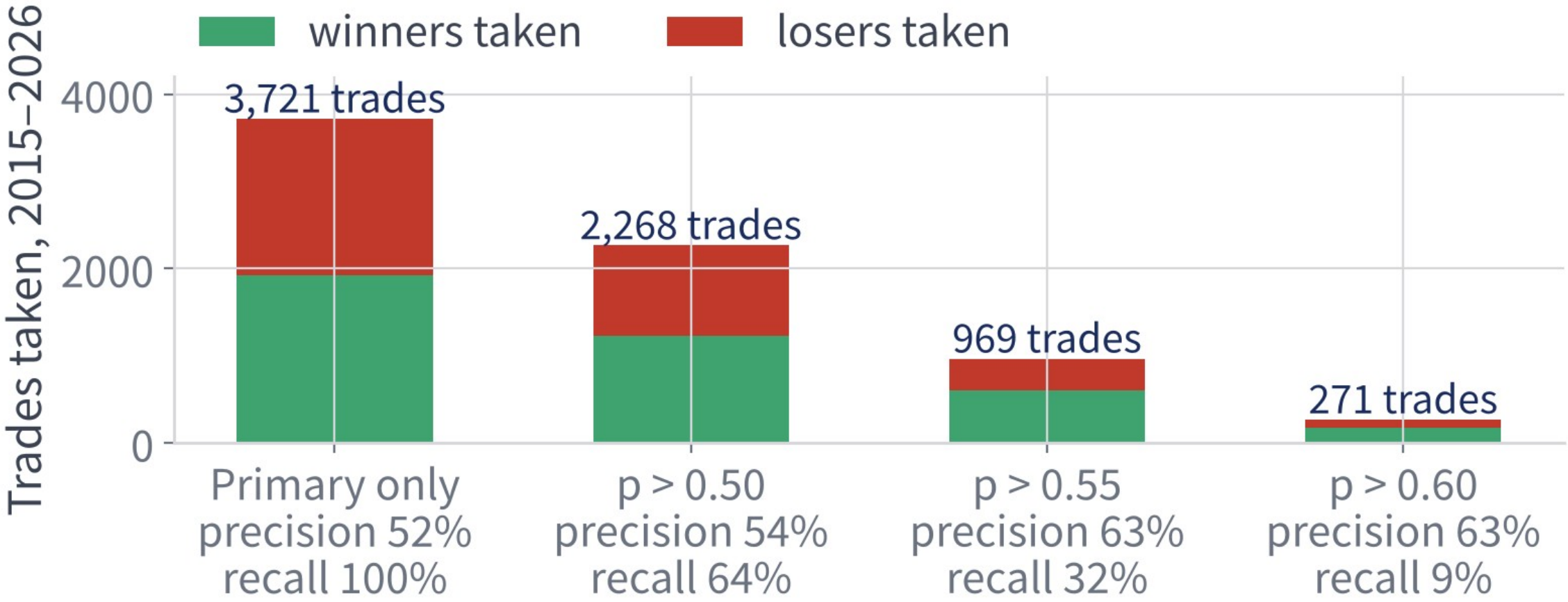
**The same 12 SPY signals after the secondary (trade only if p > 0.55)**

| | | TRADED | SKIPPED | |
|---|---|---|---|---|
| **Primary alone**<br><br>precision **50%**<br>recall 100% | would win | **TP = 4**<br>kept winners | **FN = 2**<br>missed winners | **Precision**<br>**4/4 = 100%** |
| | would lose | **FP = 0**<br>kept losers | **TN = 6**<br>avoided losers | **Recall**<br>**4/6 = 67%** |

All eight sells fought the uptrend and got low p; the four buys with trend and fear got high p

# A higher threshold keeps fewer but better trades

Precision = winners ÷ trades taken; recall = winners taken ÷ all 1,927 winners (22 ETFs, 2015–2026)

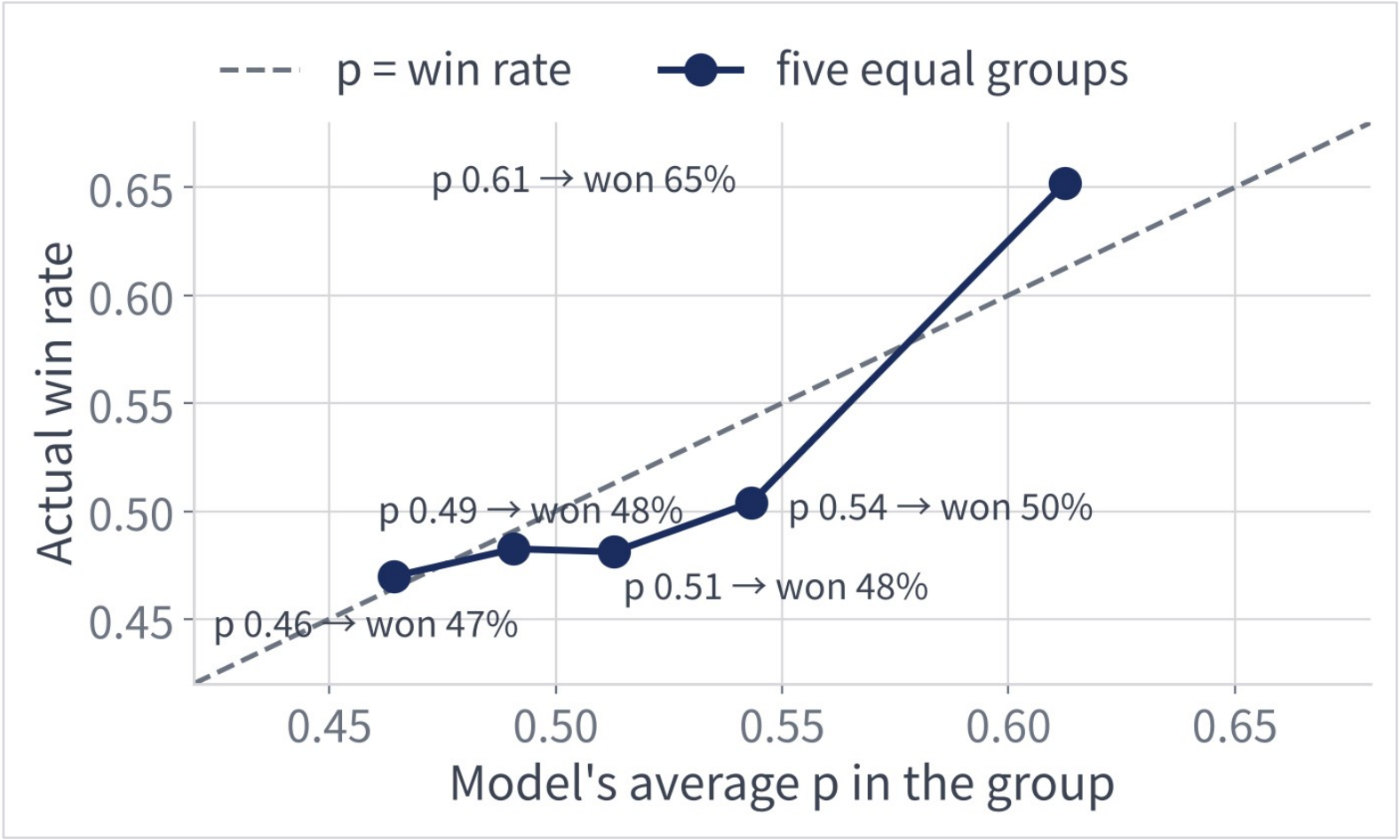


# Higher p means more wins, but not point for point

3,721 signals in five equal groups by p; on the dashed line the win rate would equal p

## What it means

- Top group: 65%
- Middle: 3–4 pts low
- p ranks well
- So bet half Kelly



03

STEP 3 · KELLY

# Kelly Turns p into Position Size

The secondary's p and the barrier odds β decide how much capital each trade gets

# Kelly takes p from the model and β from the barriers

f* is the share of capital to bet that maximizes long-run growth (Kelly, 1956; Thorp)

$$f^* = \frac{\beta \overbrace{p}^{\text{secondary}} - \overbrace{q}^{1-p}}{\underbrace{\beta}_{\text{odds from the barriers}}}$$

$$\beta = \frac{\text{average winning trade}}{\text{average losing trade}} = \frac{2.56\%}{2.32\%} = 1.10$$

### p ← secondary

- Chance this signal wins
- q = 1 – p: chance it fails

### β ← barriers (odds)

- A win pays 1.10 per 1.00 lost
- 3,006 training trades

### s ← primary

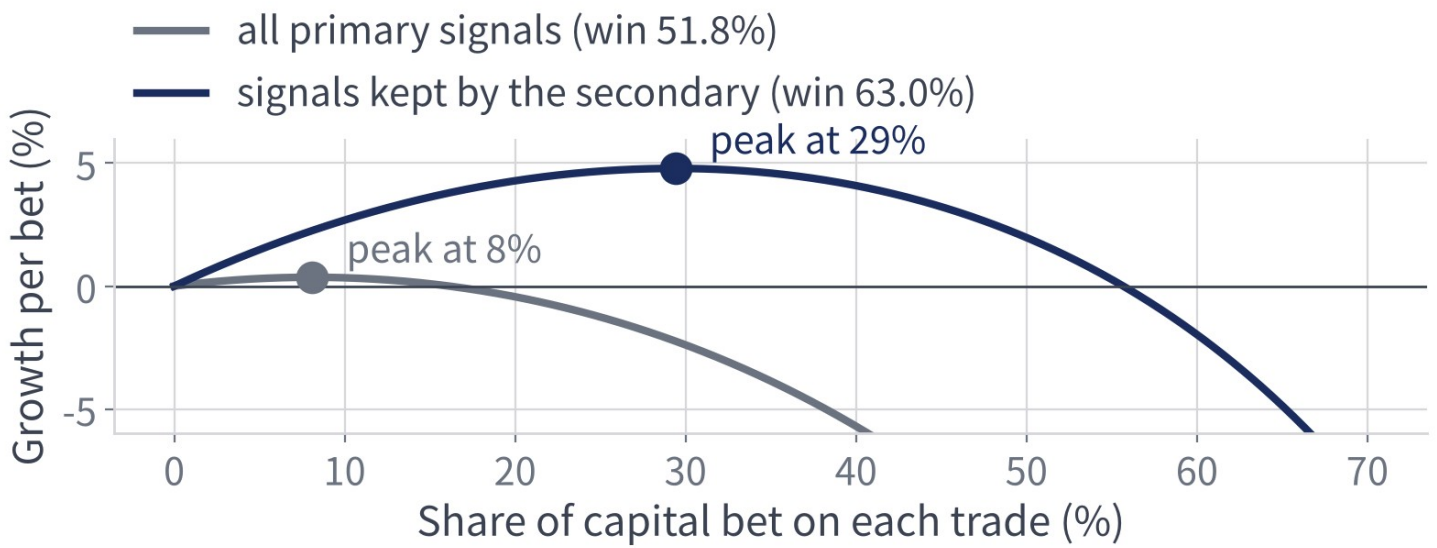
- Kelly sets only the size
- The sign sets buy or sell

# Higher precision lets Kelly bet more

Each trade replayed as a bet: a win pays 1.10× the stake, a loss costs the stake; each curve peaks at f*

$$\text{all signals (win 51.8\%):} \quad f^* = \frac{1.10 \times 0.518 - 0.482}{1.10} = 8\%$$

$$\text{kept signals (win 63.0\%):} \quad f^* = \frac{1.10 \times 0.630 - 0.370}{1.10} = 29\%$$




## Reading it

- Same odds β = 1.10
- Win rate 52% → peak 8%
- Win rate 63% → peak 29%
- Past the peak, bigger bets grow less

# One formula gives 29%, 24% and 25.4%

Same formula, three inputs: two averages over the 969 kept trades, and one single trade

| Input p | Where this p comes from | f* | ½ f* |
|---|---|---|---|
| 0.630 | Share of the 969 kept trades that won | 29% | 15% |
| 0.599 | Average of the model's p over the 969 kept trades | 24% | 12% |
| 0.609 | Model's p for the SPY buy, Nov 20 2025 | 25.4% | 12.7% |

## Why 29% is not the bet

- It uses outcomes we learn only after trading
- The model's p is lower: 0.60, not 0.63

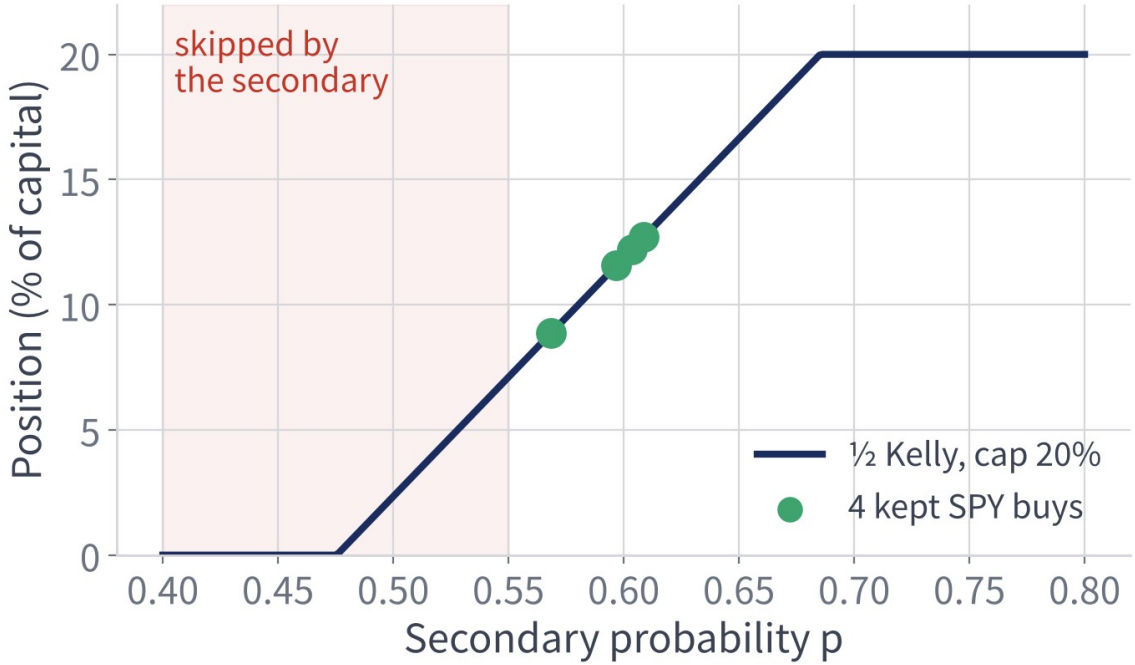
## What we actually bet

- ½ f* from each trade's own p
- Range 7–20%, average 11%

# The SPY buy of Nov 20 gets 12.7% of capital

Its own p = 0.61 gives f* = 25.4%; half Kelly guards against errors in p and a 20% cap limits any trade

$$f^* = \frac{1.10 \times 0.61 - 0.39}{1.10} = 25.4\% \quad \Rightarrow \quad \tfrac{1}{2} f^* = 12.7\% \quad \Rightarrow \quad \text{buy } 12.7\% \text{ of capital}$$

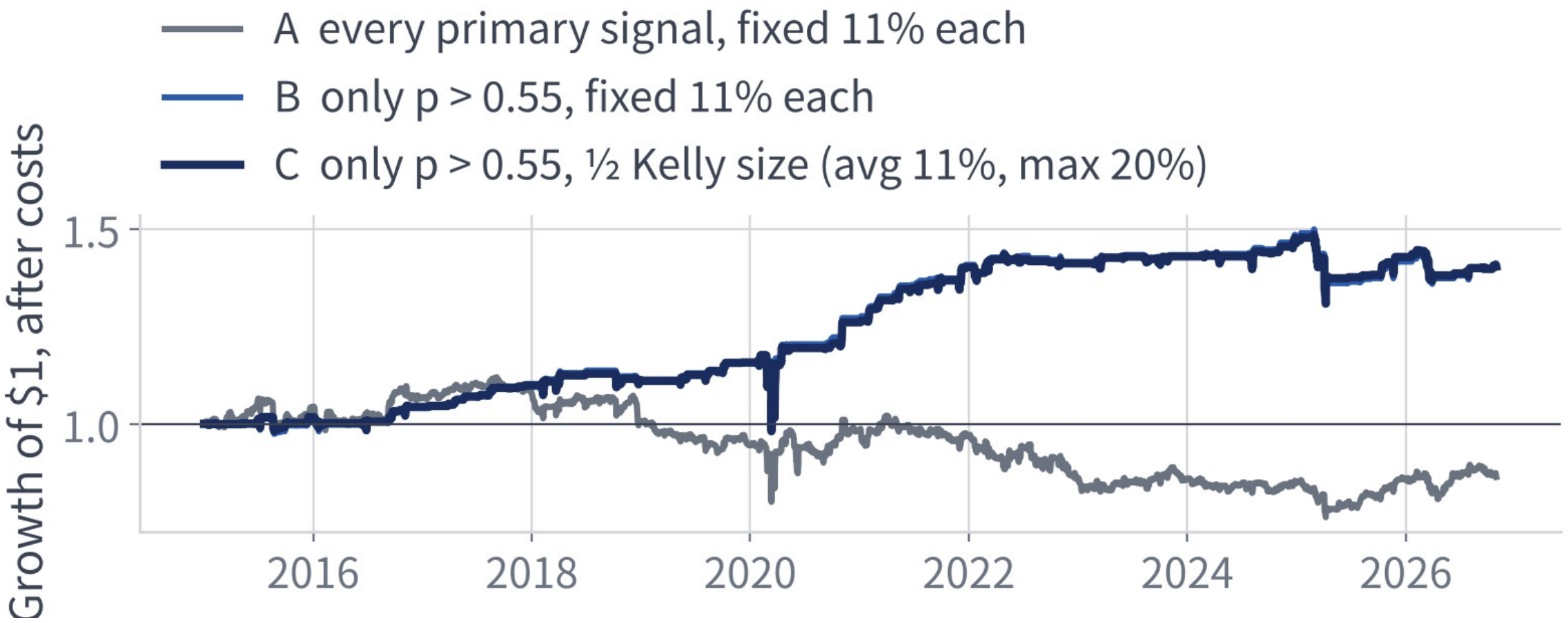



## The four kept SPY buys

- p between 0.57 and 0.61
- Sizes between 9% and 13%
- All four hit the profit line

# The secondary turns a losing rule into a winning one

A and B bet a fixed 11% per trade: the average ½ Kelly size of C's 969 trades



# Adding the secondary halves the drawdown

A → B: the filter cuts drawdown from 32% to 15%. B → C: Kelly sizing adds nothing in this sample

$$\text{fixed size of A and B} = \frac{1}{969}\sum_{i=1}^{969}\min\left(\frac{1}{2}f_i^*,\ 20\%\right) = 11\%$$

| | A All signals | B Filter only | C Filter + ½ Kelly |
|---|---|---|---|
| Which signals | all 3,721 | p > 0.55: 969 | p > 0.55: 969 |
| Size per trade | 11% fixed | 11% fixed | own ½ f*: 7–20% |
| Win rate | 51.8% | 63.0% | 63.0% |
| Sharpe ratio | −0.07 | 0.41 | 0.41 |
| Maximum drawdown | −32.1% | −14.7% | −16.6% |

C sizes each trade by its own p: SPY Nov 20 got 12.7%, median 10%, 103 trades at the 20% cap

# C fell further because p was too high in the 2020 crash

Why B → C made drawdown slightly worse: both hit their worst point between Feb 5 and Mar 12, 2020

**3.7×**
VIX vs its 60-day average (training max: 2.4×)

**0.71**
model's average p for the 58 kept signals

**48%**
of those signals actually won

**19%**
C's average size per trade (B: 11%)

## What went wrong

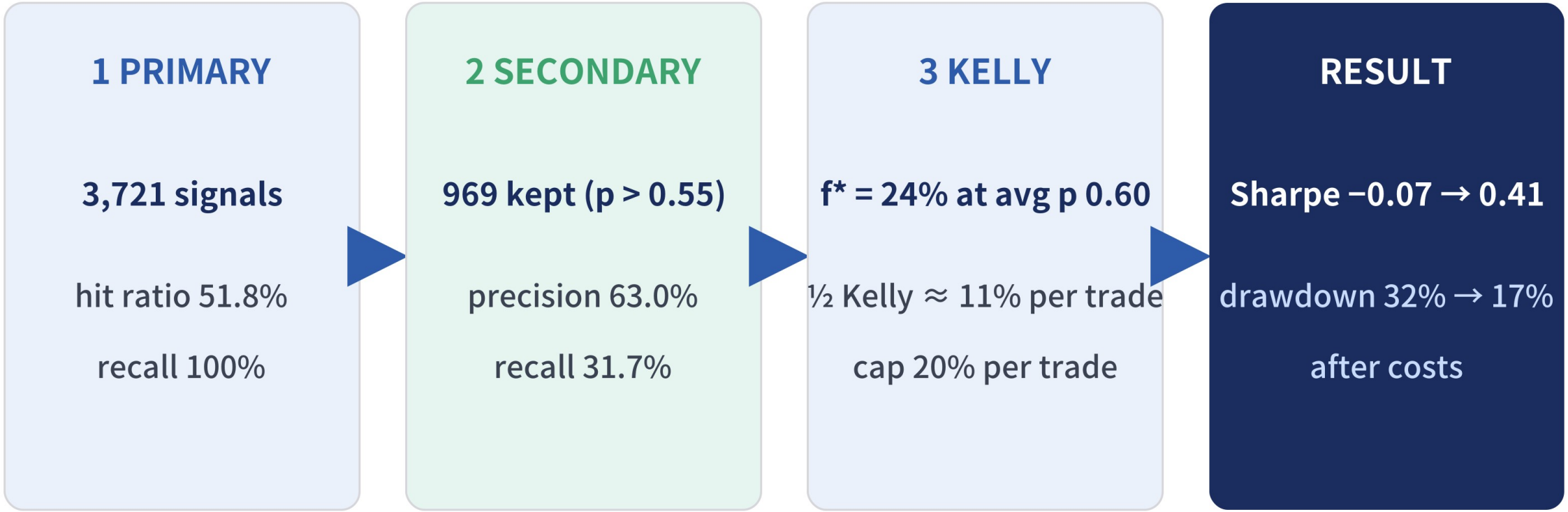
- The fear feature left its training range
- The logistic line kept raising p
- Kelly then bet almost twice as much

## How to guard against it

- Clip features to the training range
- Keep half Kelly and a size cap
- Recheck p against outcomes regularly

# Each model does one job in the pipeline

Primary for recall, secondary for precision, Kelly for size — 22 ETFs out of sample, 2015–2026



# Three lessons from the lab

1

## Let the primary be loose

Its hit ratio was only 52%, but it caught every stretched move. Missing chances is the primary's failure.

2

## Give the secondary features with a story

Trend agreement and a fear spike lifted precision from 52% to 63% with two transparent weights.

3

## Feed Kelly an honest p and the barrier odds

In the 2020 crash p averaged 0.71 but only 48% won; half Kelly and the 20% cap limited the damage.

**The primary picks the side, the secondary the odds, Kelly the size**

04

CLAUDE CODE LAB

# Build the Lab with Claude Code

Desktop app, Code tab: you paste the prompts and check the results; Claude does the rest

# Lab 1: primary signals, labels and hit ratio

Where: Claude desktop app → Code tab → new session in an empty folder

**[Input] Prompt · Claude Code (Code tab)**

```
Build a meta-labeling lab in this folder.
1. Install yfinance, pandas, scikit-learn, matplotlib.
2. Download daily closes 2004-2026 for SPY QQQ IWM DIA
   EFA EEM EWJ EWG EWU FXI XLF XLK XLE XLV XLI XLY XLP
   XLU XLB GLD TLT IEF and ^VIX.
3. Primary: Bollinger band (20d, 2 sigma). Buy when price
   crosses below -2 sigma, sell when it crosses above +2.
4. Triple barrier: +/-1.5 x daily vol, 10-day limit.
   Label 1 if the trade made money in the signal's side.
5. Print the hit ratio, list SPY signals Jun 2025-Jan 2026
   with labels, and repeat for bands 1.0/1.5/2.0/2.5.
```

**[Output] What you should see**

```
SPY: 12 signals,
  6 labeled 1 (50%)

all ETFs 2015+:
  3,721 signals
  hit ratio 51.8%

band sweep:
  wider band, more
  winners, lower hit
```

# Lab 2: logistic secondary and precision–recall

Same session: Claude keeps the files from Lab 1

**[Input] Prompt · Claude Code (Code tab)**

```
Add the secondary model.
1. Features per signal: x1 = % distance from the 200-day
   average times side; x2 = VIX / its 60-day average.
2. Train a logistic regression on signals whose labels
   ended before 2015; test on 2015-2026.
3. Print the intercept and weights. Show z and p for
   SPY on 2025-11-20 and 2025-06-24 step by step.
4. For thresholds 0.46 to 0.65 plot precision and
   recall. Show the confusion matrix at 0.55, for SPY
   Jun 2025-Jan 2026 and for all ETFs.
```

**[Output] What you should see**

```
z = -0.447
  + 0.012 x1 + 0.541 x2

SPY Nov 20: p = 0.61
SPY Jun 24: p = 0.47

at 0.55: precision 63%,
  recall 32%
```

# Lab 3: Kelly sizing, backtest and your own levers

Ask for a comparison table after every change; keep the original run as the baseline

**[Input] Prompt · Claude Code (Code tab)**

```
1. beta = avg win / avg loss on training trades.
   f* = (beta*p - (1-p))/beta; size = min(f*/2, 20%).
2. Daily backtest, 5 bp per side: A all signals at 11%,
   B p > 0.55 at 11%, C p > 0.55 sized by half Kelly.
3. Rerun with: threshold 0.50 and 0.60; band 1.5 sigma;
   full Kelly. Add one row per run to a table.
```

## Rule of the lab

- Never tune on the test years, then call it out of sample

## What to look for

- Which change moves precision?
- Which change moves only the size?
- When does full Kelly hurt?

# References and data

## Sources


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- Meyer, M., Barziy, I., Joubert, J. F. (2023). Meta-Labeling: Calibration and Position Sizing. JFDS 5(2)
- Kelly, J. L. (1956). A New Interpretation of Information Rate. Bell System Technical Journal 35(4)
- Thorp, E. O. The Kelly Criterion in Blackjack, Sports Betting, and the Stock Market
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